5 Data Exploration

Descriptive statistics

```r
#This is the definitive API dataset
efteling_live_queue <- read_csv("../../data/final/efteling_live_queue.csv")

#This is the definitive web scraped dataset with feature engineering
efteling_his_complete <- read_csv("../../data/final/efteling_his_complete.csv")
```

Historical data

```r
#Summarise definitive datasets
summary(efteling_live_queue)
```

```
##    timestamp                    attraction_name       status
##  Min.   :2025-11-13 16:04:13   Length:8215        Length:8215
##  1st Qu.:2025-11-16 10:37:58   Class :character   Class :character
##  Median :2025-11-19 12:37:39   Mode  :character   Mode  :character
##  Mean   :2025-11-20 01:28:41
##  3rd Qu.:2025-11-24 12:49:49
##  Max.   :2025-11-27 17:46:51
##
##   queue_type          wait_time
##  Length:8215        Min.   : 0.000
##  Class :character   1st Qu.: 0.000
##  Mode  :character   Median : 5.000
##                     Mean   : 9.251
##                     3rd Qu.:15.000
##                     Max.   :65.000
##                     NA's   :1319
```

The summary of the efteling_live_queue dataset shows that it contains 8,215 observations, with waiting times ranging from 0 (min. wait_time) to 65 minutes (max. wait_time). The median waiting time is 5 minutes, which indicates that at least half of the reported queues were very short. The mean is 9.251, which is slightly higher, because of some outliers of longer waiting time. The dataset includes 1,319 missing values within the wait_time variable. This means careful treatment of missing data in subsequent analysis is important.

```
## # A tibble: 12 x 5
##    month     avg_crowd_percent avg_temp_actual avg_rain_actual avg_wind_actual
##    <ord>                 <dbl>           <dbl>           <dbl>           <dbl>
##  1 januari                34.3            4.87           0.126            4.87
```

```
##  2 februari                 28.2           7.53           0.0887             4.67
##  3 maart                    35.6          10.4            0.101              4.67
##  4 april                    50.9          13.1            0.0918             4.61
##  5 mei                      51.4          17.6            0.133              4.23
##  6 juni                     57.0          21.4            0.057              4.27
##  7 juli                     45.8          21.2            0.105              3.89
##  8 augustus                 59.8          21.5            0.0538             3.64
##  9 september                41.3          19.2            0.107              4.11
## 10 oktober                  50.0          14.1            0.103              3.81
## 11 november                 40.1           9.20           0.127              4.14
## 12 december                 45.0           6.82           0.149              4.82
```

The monthly aggregate of the historical dataset indicates a seasonal pattern in both the crowd levels and the weather conditions. The data shows that average crowd percentages are lowest during winter months, such as January and February, and this percentage steadily increases towards the summer, with peaks in June and August. These peaks in crowd percentages coincide with higher average temperatures (avg_temp_actual), which indicates that warmer weather is associated with a higher number of park visitors. Rainfall levels (avg_rain_actual) does not show a strong seasonal pattern, although somewhat greater values are observed in the spring and fall. Wind conditions (avg_wind_actual) remain relatively stable during the year. Overall, these results indicate that seasonal weather conditions are correlated with fluctuations in park crowding.

```
## # A tibble: 12 x 3
##    month     avg_crowd_percent avg_queue_time
##    <ord>                 <dbl>          <dbl>
##  1 januari                40.4           14.3
##  2 februari               30.6           11.6
##  3 maart                  37.8           13.2
##  4 april                  51.4           15.9
##  5 mei                    51.9           16.4
##  6 juni                   56.6           17.2
##  7 juli                   45.9           14.8
##  8 augustus               60.3           18.0
##  9 september              42.5           14.5
## 10 oktober                50.9           15.6
## 11 november               43.1           15.8
## 12 december               48.8           16.8
```

Where in the previous table the focus was laid upon the monthly crowd levels in relation to weather conditions, this summary examines the relation between crowd levels and queue times.

The monthly averages reveal a seasonal pattern in crowd levels and queue times. Avg_crowd_percentages rise during the spring and summer months, with peaks in June and August. The increase in park attendance coincide with longer avg_queue_times. In contrast, lower crowd levels are noticeable in winter months, such as January and February, which correspond with shorter waiting times. Overall, these results indicate a strong positive correlation between the monthly crowd percentages and avg_queue_times, which indicates that periods that are busier consistently lead to longer queues in the park.

Live data

```r
#Summarise definitive datasets
summary(efteling_his_complete)
```

```
##       date               park_id        ride            avg_queue_min
##  Min.   :2023-01-01   Min.   :160   Length:31342       Min.   :  5.00
##  1st Qu.:2023-09-28   1st Qu.:160   Class :character   1st Qu.:  7.00
##  Median :2024-06-24   Median :160   Mode  :character   Median : 13.00
##  Mean   :2024-06-17   Mean   :160                      Mean   : 15.44
##  3rd Qu.:2025-03-09   3rd Qu.:160                      3rd Qu.: 22.00
##  Max.   :2025-11-16   Max.   :160                      Max.   :141.00
##                                                        NA's   :5290
##  max_queue_min      uptime_pct     crowd_percent    crowd_label
##  Min.   :  5.00   Min.   :  0.00   Min.   :  0.00   Length:31342
##  1st Qu.: 15.00   1st Qu.: 89.00   1st Qu.: 23.00   Class :character
##  Median : 25.00   Median : 99.00   Median : 43.00   Mode  :character
##  Mean   : 26.36   Mean   : 82.91   Mean   : 45.13
##  3rd Qu.: 35.00   3rd Qu.:100.00   3rd Qu.: 66.00
##  Max.   :260.00   Max.   :100.00   Max.   :100.00
##  NA's   :5290
##  temperature_forecast temperature_actual rain_forecast     rain_actual
##  Min.   :-2.80        Min.   :-1.70      Min.   :0.0000   Min.   :0.0000
##  1st Qu.: 8.80        1st Qu.: 9.20      1st Qu.:0.0000   1st Qu.:0.0000
##  Median :14.30        Median :14.10      Median :0.0000   Median :0.0000
##  Mean   :14.28        Mean   :14.22      Mean   :0.1023   Mean   :0.1011
##  3rd Qu.:20.00        3rd Qu.:19.50      3rd Qu.:0.1000   3rd Qu.:0.0800
##  Max.   :34.10        Max.   :34.80      Max.   :2.0900   Max.   :2.2300
##  NA's   :2069                            NA's   :2069
##  wind_forecast     wind_actual        events           weekday_weekend
##  Min.   : 0.500   Min.   : 0.900   Length:31342       Length:31342
##  1st Qu.: 2.500   1st Qu.: 3.000   Class :character   Class :character
##  Median : 3.500   Median : 4.100   Mode  :character   Mode  :character
##  Mean   : 3.672   Mean   : 4.288
##  3rd Qu.: 4.600   3rd Qu.: 5.300
##  Max.   :11.200   Max.   :12.400
##  NA's   :2069
##  has_single_rider indoor_outdoor     capacity_theoretical  opening_year
##  Min.   :0.0000   Length:31342       Min.   :  50         Min.   :1956
##  1st Qu.:0.0000   Class :character   1st Qu.:1000         1st Qu.:1982
##  Median :0.0000   Mode  :character   Median :1400         Median :1993
##  Mean   :0.2134                      Mean   :1292         Mean   :1996
##  3rd Qu.:0.0000                      3rd Qu.:1800         3rd Qu.:2015
##  Max.   :1.0000                      Max.   :2010         Max.   :2024
##
##  is_school_holiday_any is_holiday
##  Mode :logical         Mode :logical
##  FALSE:20841           FALSE:30421
##  TRUE :10501           TRUE :921
##
##
##
##
```

This definitive data set of the historical data contains 31,342 observations covering the period from January 2023 to November 2025. Average queue times range from 5 (min) to 141 (max) minutes, with 5,290 missing values. Crowd levels vary between the 0 and the 100%, which shows variability throughout the data. Weather variables, like temperature, rainfall, and wind speed vary widely across the data set. Additional contextual information relevant for modelling queue dynamics, such as ride type, or indoor vs. outdoor, are also incorporate in this data set.

```r
#Of live data, calculate average queue time
efteling_live_queue %>%
  group_by(attraction_name, queue_type) %>%
  summarise(averagequeue = mean(wait_time, na.rm = TRUE)) %>%
  arrange(desc(averagequeue)) %>%
  print(n = 31)
```

```
## `summarise()` has grouped output by 'attraction_name'. You can override using
## the `.groups` argument.
```

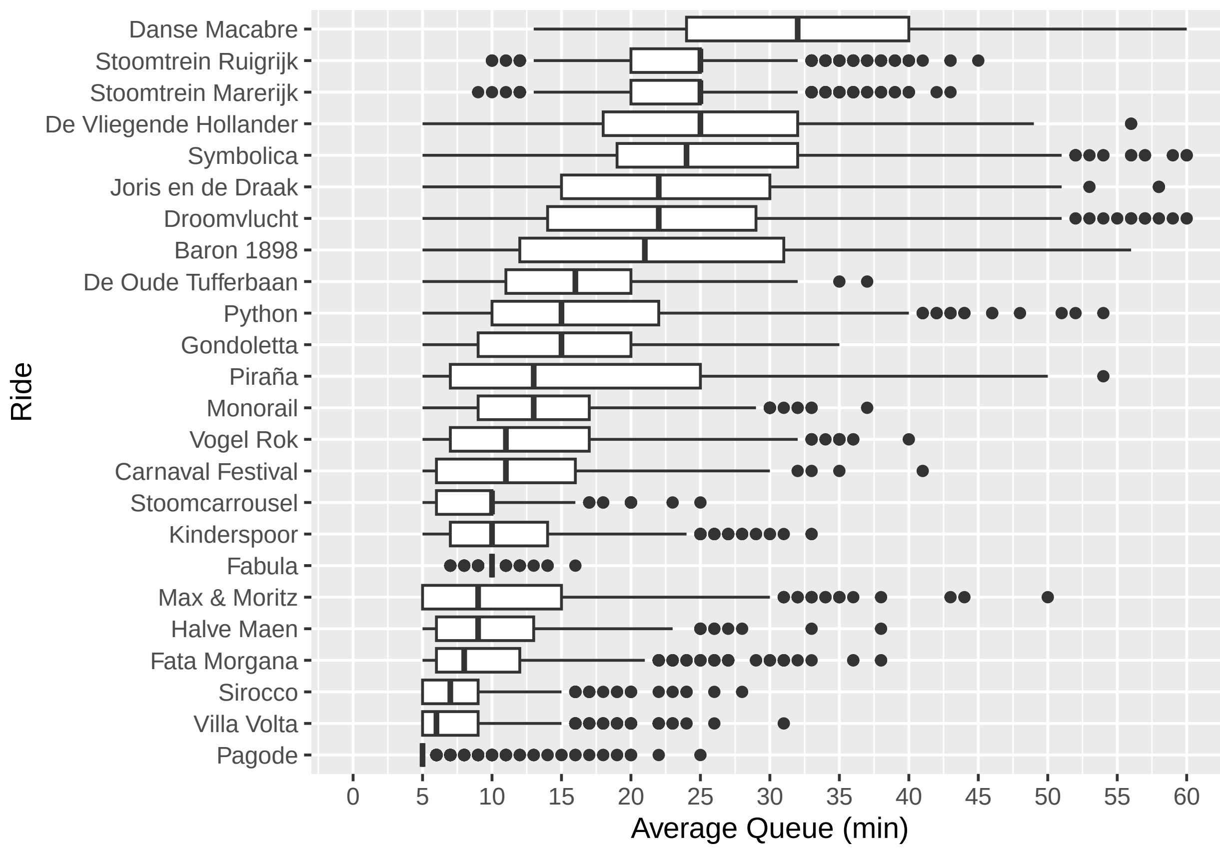
```
## # A tibble: 31 x 3
## # Groups:   attraction_name [31]
##    attraction_name                    queue_type   averagequeue
##    <chr>                              <chr>               <dbl>
##  1 Danse Macabre                      STANDBY            25.7
##  2 Stoomtrein Ruigrijk                STANDBY            23.8
##  3 Stoomtrein Marerijk                STANDBY            23.7
##  4 Symbolica                          STANDBY            23.1
##  5 Baron 1898                         STANDBY            17.4
##  6 Baron 1898 Single-rider            SINGLE_RIDER       15.6
##  7 Joris en de Draak                  STANDBY            15.0
##  8 Droomvlucht                        STANDBY            14.7
##  9 Python                             STANDBY            12.1
## 10 De Vliegende Hollander             STANDBY            11.9
## 11 Fabula                             STANDBY             9.98
## 12 De Oude Tufferbaan                 STANDBY             9.42
## 13 Max & Moritz                       STANDBY             9.18
## 14 Vogel Rok                          STANDBY             7.68
## 15 Fata Morgana                       STANDBY             7.09
## 16 Carnaval Festival                  STANDBY             6.31
## 17 Villa Volta                        STANDBY             5.55
## 18 Halve Maen                         STANDBY             5.12
## 19 Pagode                             STANDBY             4.94
## 20 Joris en de Draak Single-rider     SINGLE_RIDER        4.32
## 21 Kinderspoor                        STANDBY             2.95
## 22 Monorail                           STANDBY             2.94
## 23 Max & Moritz Single-rider          SINGLE_RIDER        2.87
## 24 Symbolica Single-rider             SINGLE_RIDER        2.84
## 25 Piraña                             STANDBY             2.53
## 26 Python Single-rider                SINGLE_RIDER        2.16
## 27 Gondoletta                         STANDBY             1.92
## 28 De Vliegende Hollander Single-rider SINGLE_RIDER       0.822
## 29 Stoomcarrousel                     STANDBY             0.797
## 30 Danse Macabre Single-rider         SINGLE_RIDER        0
## 31 Sirocco                            STANDBY           NaN
```

Utilizing the live queue data set, which contains 8,215 observations, the analysis identified all 31 attractions of which the average waiting time can be calculated. The table shows all the attractions in combination with their corresponding queue type, such as standby or single_rider. This output can be used to compare queue dynamics across rides and queue systems.

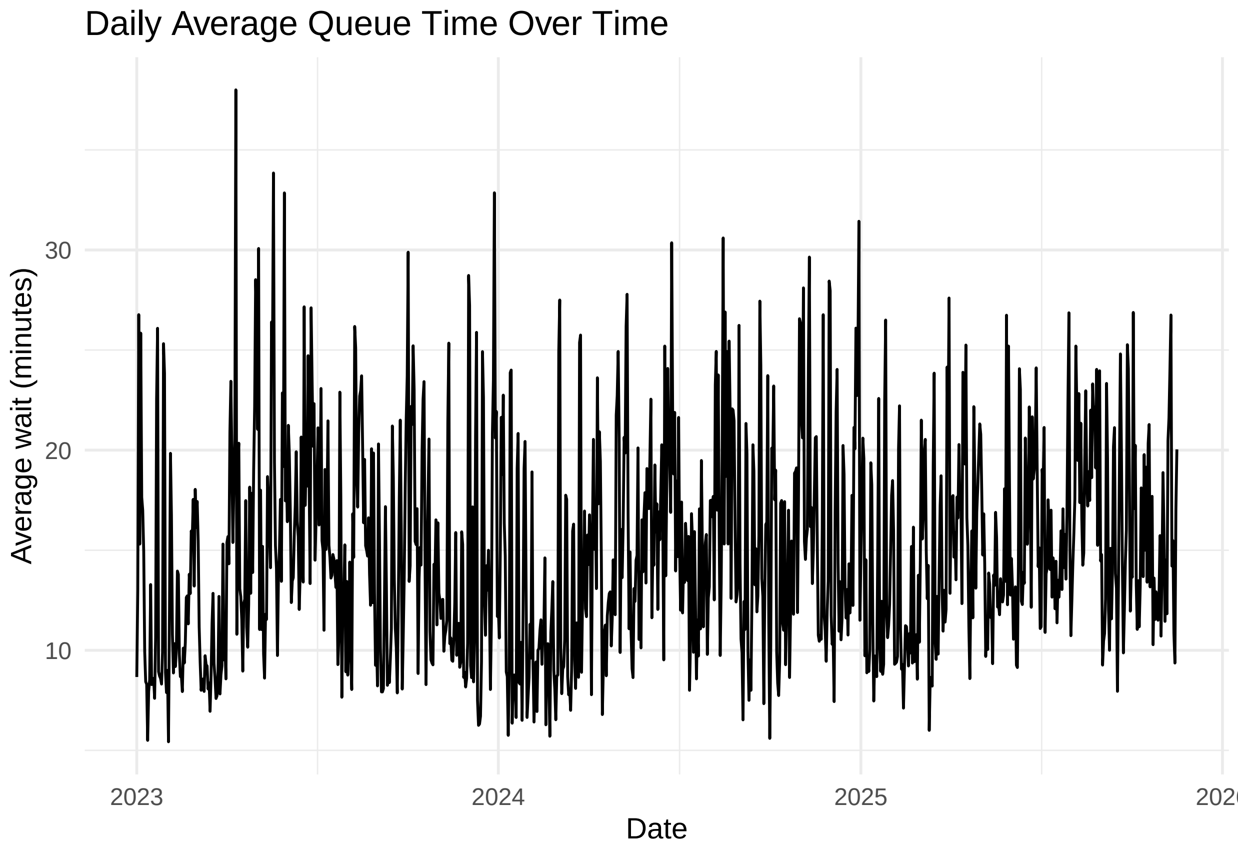
Data Visualisations

Historical data

```
## Warning: Removed 40 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

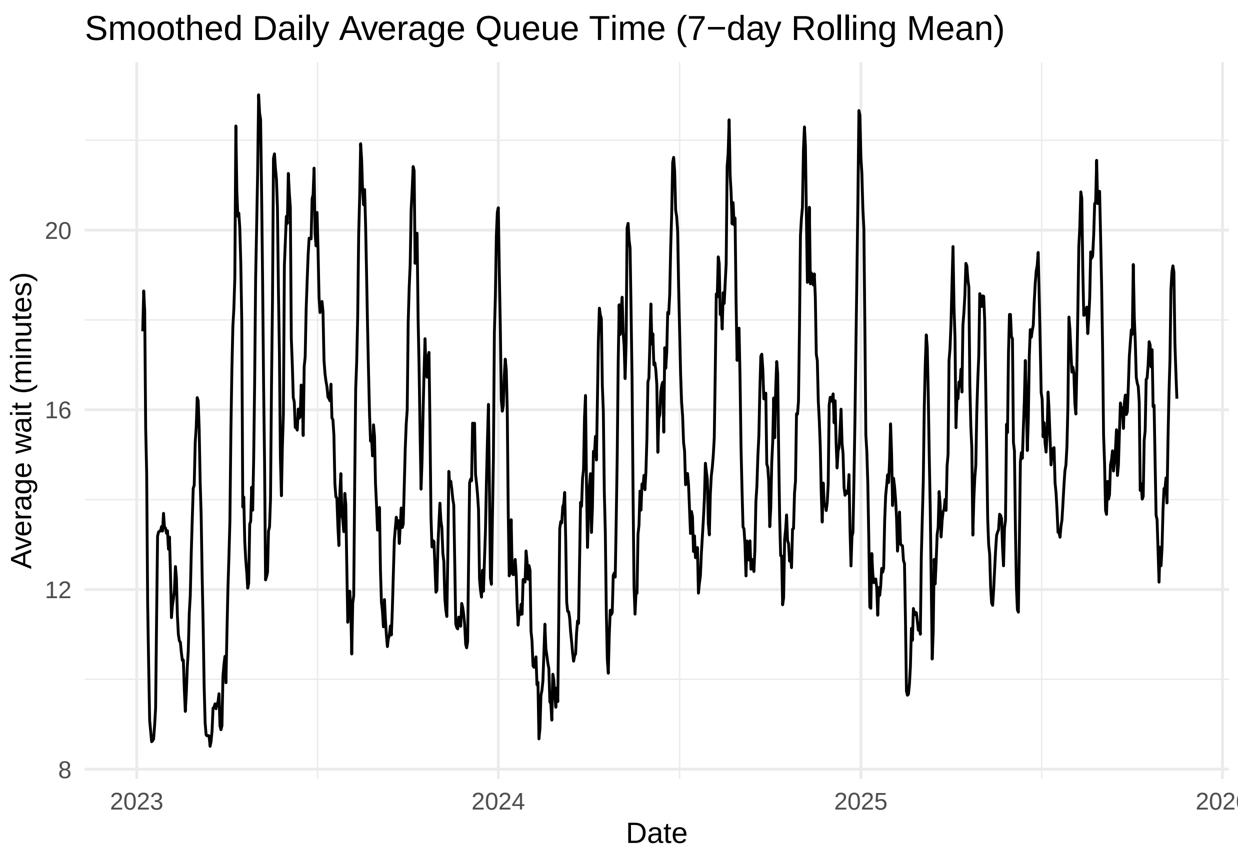


This boxplot shows a summary of the historical queue times for the 31 attractions, with an exclusion on the single-rider lines. A big part of the rides show median waiting times between the 10 and 30 minutes. Several rides, such as Symbolica, Joris en de Draak, and Baron 1898, show variability and occasional peaks above the 40 minutes. The x-axis was limited to 60 minutes for readability, and two rides has a small number of exterme outliers, that ranged above the 60 minutes. Overall, the boxplot shows the differences in queue times across the rides, where extreme rides generally show longer and more fluctuating wait times.

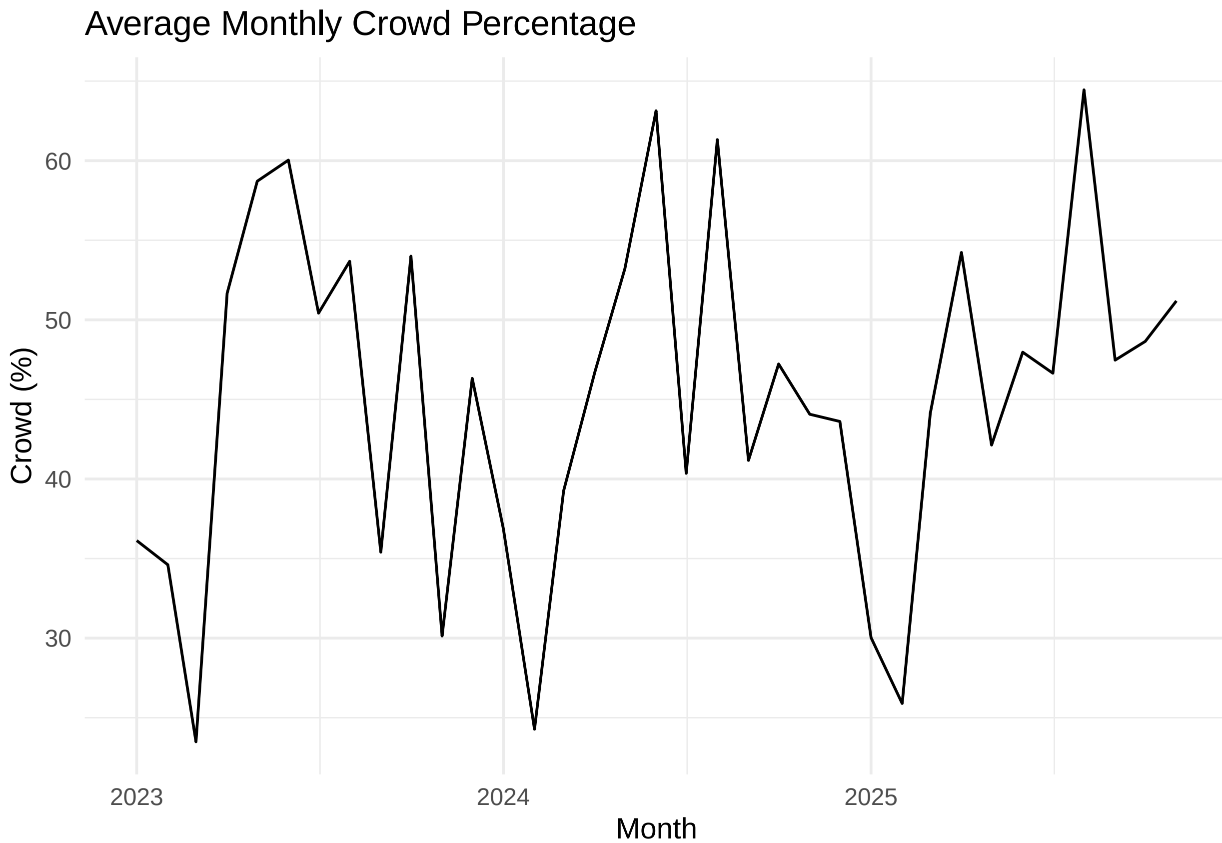


This time-series plot shows the daily average queue time across all attractions. The data fluctuates strongly from day to day, reflecting natural variation due to weather, crowds, holidays, and operational factors. Several clear peaks occur during busy periods, while dips reflect lower-attendance days. Because the raw data is unsmoothed, the figure emphasises how volatile daily queue patterns can be.

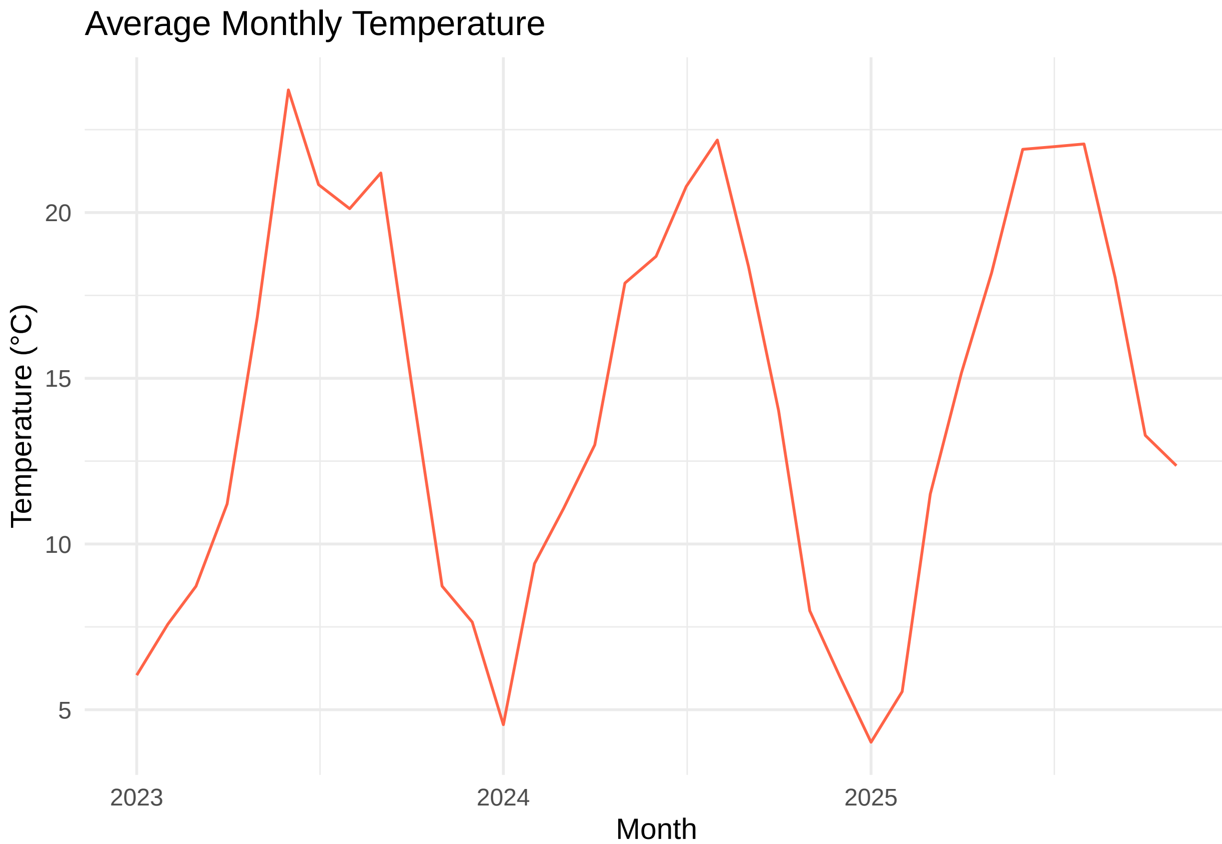
```
## Warning: Removed 6 rows containing missing values or values outside the scale
## range (‘geom_line()‘).
```



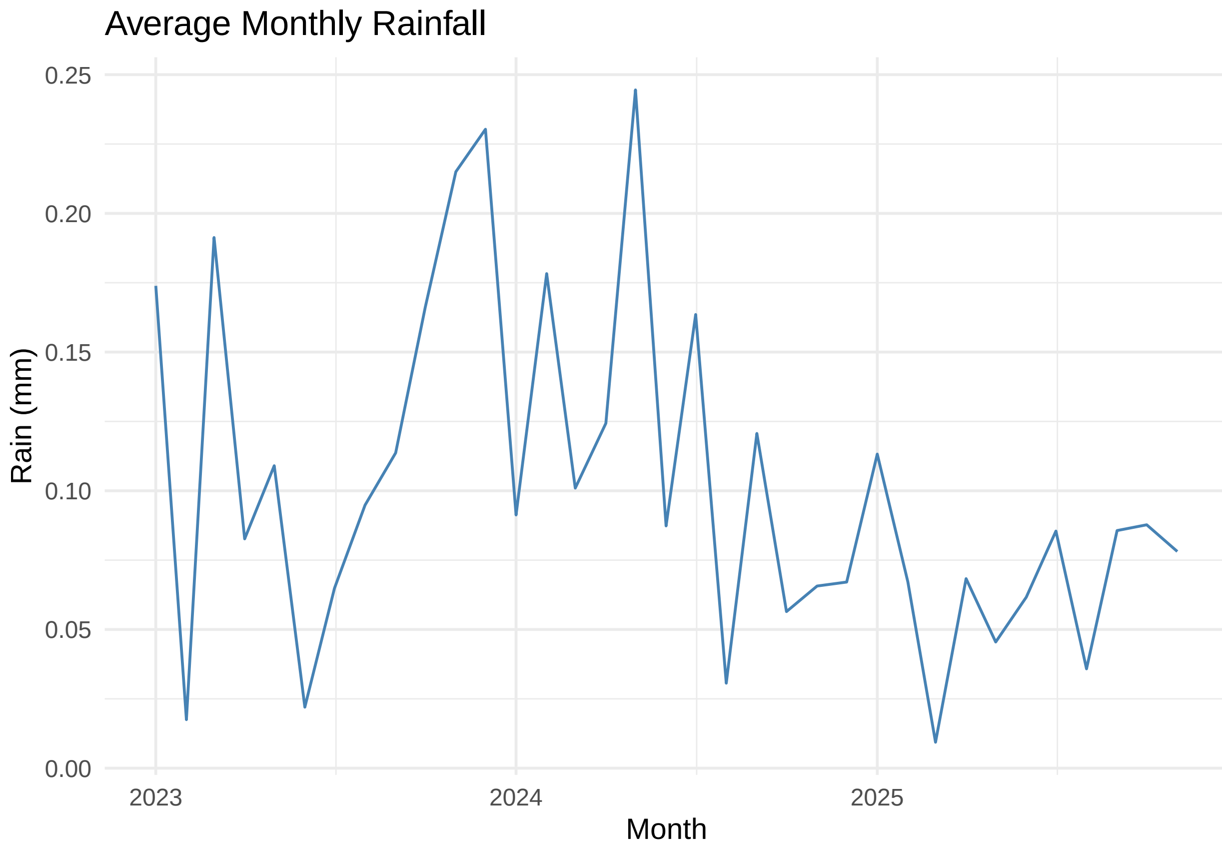
This figure shows the 7-day rolling average of daily queue times, providing a smoother representation of the trend. Short-term fluctuations are reduced, making recurring seasonal patterns easier to observe. Peaks align with known high-attendance periods, such as summer holidays or special events. Compared to the raw daily series, this plot gives a clearer overview of underlying trends in visitor behaviour.



This plot shows the average monthly crowd level across the historical dataset. Crowd percentages rise sharply during spring and summer, particularly around school holidays, and decline during the winter months. These patterns reflect typical theme-park attendance cycles. The monthly view highlights how predictable seasonality influences the general occupancy of the park.

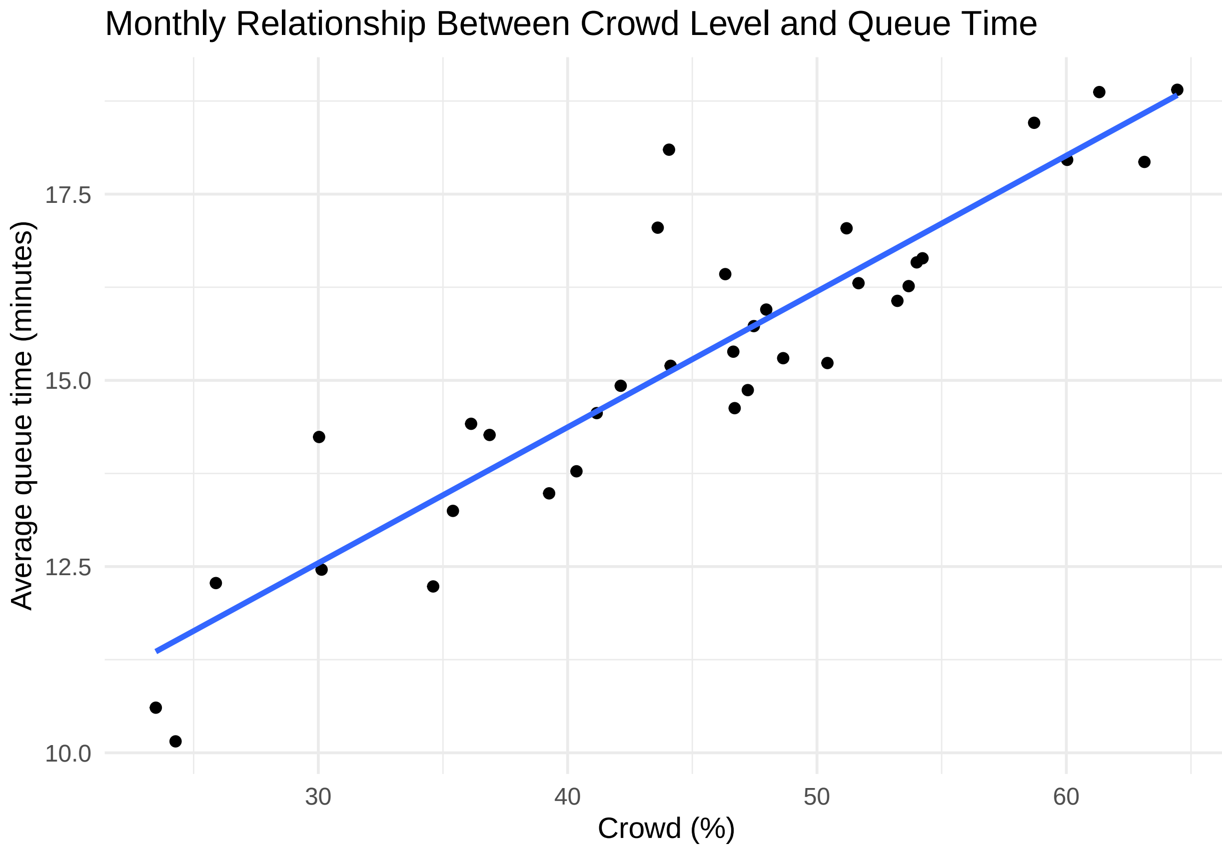


This figure presents the average monthly temperature throughout the year. Temperatures peak during June–August, while the coldest months appear in winter. The seasonal pattern helps contextualise visitor behaviour, as weather conditions can influence both attendance and queue times. Overall, this variable aligns with expected climate trends.

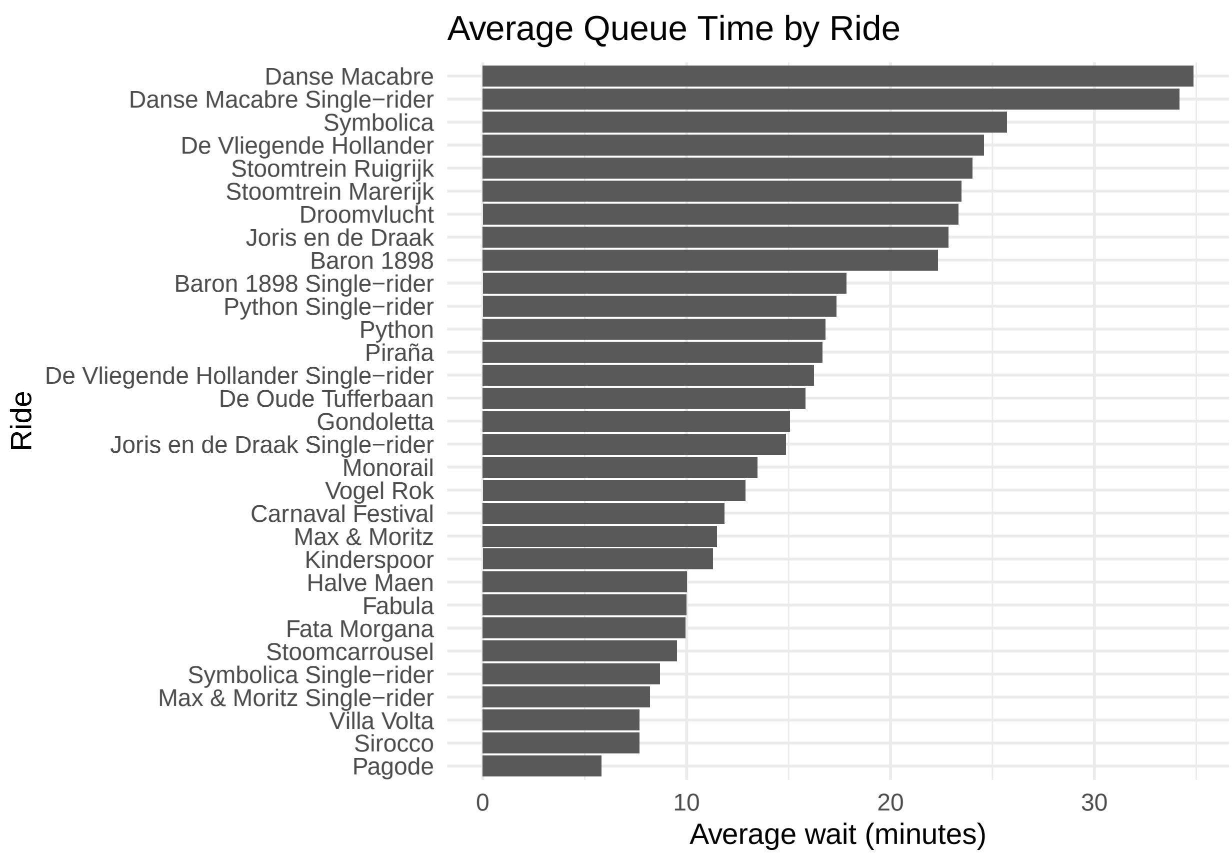


This figure shows the average rainfall per month. Rainfall appears to vary substantially across the year, with wetter months typically occurring in autumn and early spring. Higher rainfall months may correspond to lower attendance levels, although the exact effect varies across days and ride types. The plot helps illustrate how weather conditions fluctuate across seasons.

```
## `geom_smooth()` using formula = 'y ~ x'
```



This scatterplot shows the relationship between monthly crowd percentages and average monthly queue times. The clear upward trend indicates that higher crowd levels correspond to longer waiting times. Months with low crowd percentages generally exhibit shorter queues, while highly attended months show substantially longer averages. The fitted regression line emphasises the strength of this positive relationship.



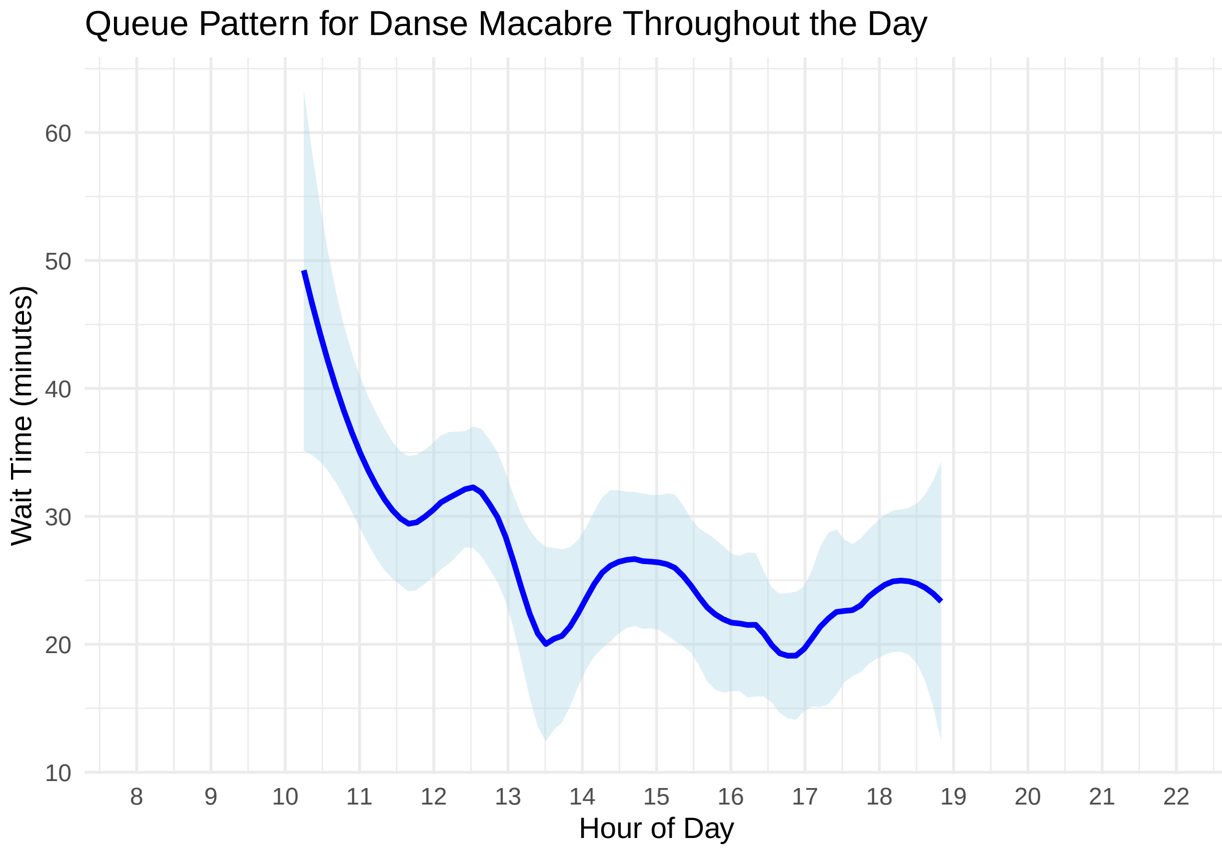
This bar chart shows the average queue time for each attraction across the historical dataset. Some rides, such as Baron 1898, Symbolica, and Droomvlucht, consistently have higher waiting times, indicating strong popularity combined with limited capacity. Other rides show much shorter averages, reflecting either higher throughput or lower demand. The figure provides a clear comparison of typical waiting times across all attractions.

Live data

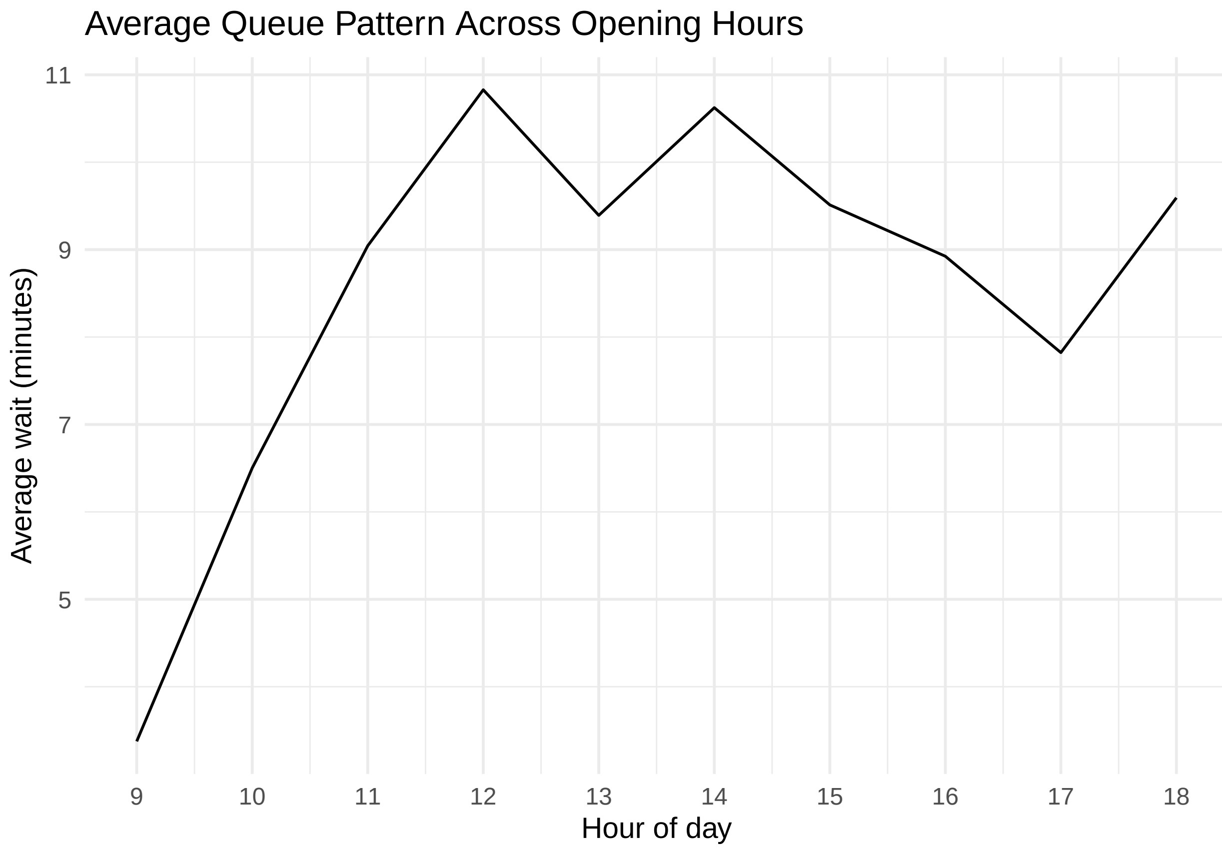
```r
#Queue pattern identification for one particular ride
efteling_live_queue %>%
  filter(attraction_name == "Danse Macabre") %>%
  filter(queue_type == "STANDBY") %>%
  mutate(hour = hour(timestamp) + minute(timestamp)/60) %>%
  ggplot(aes(x = hour, y = wait_time)) +
  geom_smooth(method = "loess", span = 0.3, se = TRUE, color = "blue", fill = "lightblue") +
  scale_x_continuous(breaks = 8:22, limits = c(8, 22)) +
  labs(
    title = "Queue Pattern for Danse Macabre Throughout the Day",
    x = "Hour of Day",
    y = "Wait Time (minutes)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 88 rows containing non-finite outside the scale range
## (‘stat_smooth()‘).
```



The Queue pattern for Danse Macabre shows a clear decline in waiting times over the day. The queue begins at its highest point around 10 am, where it is approximately 50 - 55 minuters. After this it drops towards the early afternoon. Between 1 pm and 7 pm, the wait time stabilizes at a lower level of 18 - 25 minutes. This daily pattern indicates that the morning period consistently experiences the highest crowd pressure for this attraction.



This figure shows the average live waiting time per hour across all attractions. Queue lengths increase rapidly, shortly after opening and reach their highest levels around midday and early afternoon. Towards the evening, the average waiting time gradually declines as crowds disperse or leave the park. The plot reflects the typical daily rhythm of visitor behaviour in a theme-park environment.